Dr Adam Dillamore

POSTDOCTORAL RESEARCH ASSISTANT · ASTROPHYSICS

University College London

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Positions

2024-present **Postdoctoral Research Assistant in Astrophysics**. *University College London*

Education

University of Cambridge (Churchill College) *Cambridge, UK*

PHD, ASTRONOMY *2021 - 2024*

- Thesis: Formation and dynamics of the Galactic disc and halo
- Supervisors: Prof. Vasily Belokurov & Prof. N. Wyn Evans

University of Cambridge (Churchill College) *Cambridge, UK*

MSCI & BA, NATURAL SCIENCES (1ST) *2017 - 2021*

- Master's supervisors: Prof. Vasily Belokurov, Dr. Andreea S. Font & Prof. Ian G. McCarthy

First-author publications

PUBLISHED

GSE vs. LMC: reshaping of radially biased stellar haloes by satellites. MNRAS, 2026.
Adam M. Dillamore, Richard A. N. Brooks and Jason L. Sanders
`https://doi.org/10.1093/mnras/stag1110`

Geometry of the Milky Way's dark matter from dynamical models of the tilted stellar halo. MNRAS, 2026.
Adam M. Dillamore and Jason L. Sanders
`https://doi.org/10.1093/mnras/stag226`

Bar-driven dispersal of Galactic substructure. MNRAS, 2025.
Adam M. Dillamore and Jason L. Sanders
`https://doi.org/10.1093/mnras/staf1264`

Dynamical streams in the local stellar halo. MNRAS, 2025.
Adam M. Dillamore, Jason L. Sanders, Vasily Belokurov and Hanyuan Zhang
`https://doi.org/10.1093/mnras/staf965`

Radial halo substructure in harmony with the Galactic bar. MNRAS, 2024.
Adam M. Dillamore, Vasily Belokurov and N. Wyn Evans
`https://doi.org/10.1093/mnras/stae1789`

Trojan Globular Clusters: Radial Migration via Trapping in Bar Resonances. ApJL, 2024.
Adam M. Dillamore, Stephanie Monty, Vasily Belokurov and N. Wyn Evans
`https://doi.org/10.3847/2041-8213/ad60c8`

Taking the Milky Way for a spin: disc formation in the ARTEMIS simulations. MNRAS, 2023.
Adam M. Dillamore, Vasily Belokurov, Andrey Kravtsov and Andreea S. Font
`https://doi.org/10.1093/mnras/stad3369`

Stellar halo substructure generated by bar resonances. MNRAS, 2023.
Adam M. Dillamore, Vasily Belokurov, N. Wyn Evans and Elliot Y. Davies
`https://doi.org/10.1093/mnras/stad2136`

A correlation between accreted stellar kinematics and dark-matter halo spin in the ARTEMIS simulations. MNRAS Letters, 2023.
Adam M. Dillamore, Vasily Belokurov, N. Wyn Evans and Andreea S. Font
`https://doi.org/10.1093/mnrasl/slac158`

The impact of a massive Sagittarius dSph on GD-1-like streams. MNRAS, 2022.
Adam M. Dillamore, Vasily Belokurov, N. Wyn Evans and Adrian M. Price-Whelan
`https://doi.org/10.1093/mnras/stac2311`

Merger-induced galaxy transformations in the ARTEMIS simulations. MNRAS, 2022.
Adam M. Dillamore, Vasily Belokurov, Andreea S. Font and Ian G. McCarthy
`https://doi.org/10.1093/mnras/stac1038`

Submitted

Bar-induced migration of ω Centauri away from Gaia Sausage-Enceladus. MNRAS submitted, 2026.
Adam M. Dillamore, Hanyuan Zhang and Vasily Belokurov

Awards and Prizes

2023 **Murdin Prize**, Institute of Astronomy, University of Cambridge
For the best published journal paper produced by a current PhD student.

2021-2025 **STFC PhD Studentship**, Institute of Astronomy, University of Cambridge
Funding for up to 3.5 years to complete a PhD.

2021 **Institute of Astronomy Prize**, Institute of Astronomy, University of Cambridge
Awarded annually to that candidate for Astrophysics in Part III of the Natural Sciences Tripos or a Master of Advanced Study in Astrophysics candidate who has, in the judgement of the Examiners shown the greatest distinction in that examination, provided that his or her work is of sufficient merit.

Presentations

Invited Talks

Taking the Milky Way for a spin: disc formation in the ARTEMIS simulations
National Astronomy Meeting 2024 (Jul 2024). Hull, UK

Contributed talks

Dynamical models of the Milky Way's tilted halo
National Astronomy Meeting 2026 (Jul 2026). Birmingham, UK

Dynamical models of the Milky Way's tilted halo
HOW STARS MOVE AROUND AND WHAT THEY TELL US ABOUT GALAXIES' MASSES, FORMATION AND EVOLUTION (Jul 2026). Sexten, Italy

Bar-driven dispersal of Galactic substructure
Galactic Bars, Bulges and Disks (Jun 2026). Lipari, Italy

Dynamical streams in the local stellar halo
National Astronomy Meeting 2025 (Jul 2025). Durham, UK

Tidal tails of Trojan globular clusters
Streams 24 (Aug 2024). Durham, UK

Bar resonances in the highly eccentric stellar halo
The Milky Way Assembly Tale (May 2024). Bologna, Italy

Taking the Milky Way for a spin: disc formation in the ARTEMIS simulations
The Milky Way and its high-redshift progenitors in theory and observations (Dec 2023). Cambridge, UK

Stellar halo substructure generated by bar resonances
Revealed by Gaia: the central halo of the Milky Way (Sep 2023). Cambridge, UK

A stellar halo walks into a bar: creation of substructure
European Astronomical Society Annual Meeting (Jul 2023). Kraków, Poland

A stellar halo walks into a bar: creation of substructure
Galactic bars: driving and decoding galaxy evolution (Jul 2023). Granada, Spain

Department talks

Creation and destruction of halo substructure by the Galactic bar
Surrey-London Astro Meeting (Apr 2024). UCL, London, UK

The impact of a massive Sagittarius Dwarf Galaxy on stellar streams
IoA Wednesday Seminar (Jan 2023). Institute of Astronomy, University of Cambridge, UK

Merger-induced galaxy transformations in the ARTEMIS simulations
IoA Galaxies Discussion Group (Nov 2021). Institute of Astronomy, University of Cambridge, UK

Public talks

An Archaeologist's Guide to the Galaxy
British Astronomical Association Deep Sky Section webinar (Oct 2024). Online

Teaching Experience

2021-2024 **Mathematics for Part IB Natural Sciences**, Supervisor
Churchill College, University of Cambridge. Total of ∼120 hours from 2021-2024

2021-2024 **Stellar Dynamics and Structure of Galaxies (Part II Astrophysics)**, Supervisor
Institute of Astronomy, University of Cambridge. Total of ∼30 hours from 2021-2024

Mentoring

2025-present **Alice Archer**, Part III student, Institute of Astronomy, University of Cambridge
I co-supervised a master's project entitled 'Revealing Phase-Space Chevrons with RR Lyrae: The Galactic Bar and Its Interaction with the Dark Matter Halo'. This is currently being written up for publication.

2025-2026 **Amy Ruth Tamerat**, MSci student, University College London
I co-supervised a master's project entitled 'Dynamical structure of stellar haloes in simulated Milky Way analogues'.

2022-2023 **Fred Thompson**, Part III student, Institute of Astronomy, University of Cambridge
I co-supervised a master's project entitled 'Dwarf Galaxy Interactions in the ARTEMIS Simulations'.

Software Skills

Python **NumPy, SciPy, Astropy, Matplotlib**.
Agama. *Galactic dynamics package. Includes potential construction, orbit integration, calculation of action-angle variables, distribution function generation, and Schwarzschild modelling.*
pyfalcon. *Python interface for the gravity solver falcON. Used to run N-body simulations.*
emcee. *Python implementation of Markov chain Monte Carlo (MCMC) sampling.*
h5py. *Python interface to HDF5 files. Used to access cosmological simulation snapshots.*

SQL **Gaia Archive**.
EAGLE Database. *Used to access merger trees of the ARTEMIS simulations.*
WSDB (Whole Sky DataBase). *Database server with the data from major sky surveys in SQL accessible form.*

Outreach & Widening Participation

2022 **STEM SMART**, Mentor
A widening participation initiative from the University of Cambridge to provide free, complementary teaching and support to UK students, including those who have experienced educational disadvantage.

2021-2022 **Undergraduate Journal Club**, Organiser
I organised and ran a fortnightly journal club for 3rd year undergraduates and 4th year/master's students at the Institute of Astronomy, University of Cambridge.